

On the Multiplicatively Closed Set Generated by 2 and 3

Partial results, an exact census up to 10^8 , and a rejected mean-field heuristic

Research dossier for Ben Green's open problem GREEN-036 (100 Open Problems, Number Theory) - August 25, 2026.

Source: <https://www.unsolvedmath.com/problems/GREEN-036> - status: **OPEN**

Abstract. Let A be the smallest set of positive integers containing 2 and 3 and closed under the operation $(a_1, a_2) \mapsto a_1 a_2 - 1$. Green's problem GREEN-036 asks whether A has positive asymptotic density. **The problem remains open.** This dossier contributes: (i) complete proofs of a well-foundedness structure theorem, of the modular obstruction (Lemma A: A avoids $1 \pmod 3$, whence upper density $d^*(A) \leq 2/3$), and of an explicit $\text{floor}((\text{floor}(\log_2 N) - 1)^{2/4}) \sim 0.5203 (\ln N)^2$ lower bound (Lemma B); (ii) an exact, independently cross-checked census $F(10^8) = |A \cap [1, 10^8]| = 55,939,931$ with per-decade statistics showing monotonically rising density $0.500 \rightarrow 0.559$ and local power-law exponents descending to 1.0079 ; (iii) a mean-field self-consistency heuristic $\sum_{b \in A} b^{-(1+\alpha)} = 1$ predicting $\alpha^* = 0.4734$, which is quantitatively rejected by the census (factor 9124), indicating that membership correlations, not pseudorandomness, govern A ; and (iv) a residue census showing that among all moduli $m \leq 500$ only $m = 3$ yields a proper closed residue set, so no improvement on $2/3$ is known. Statements are labeled [PROVEN], [COMPUTED], or [HEURISTIC] throughout.

1. Problem statement and status

Let A be minimal with $2, 3 \in A$ and $a_1, a_2 \in A$ implying $a_1 a_2 - 1 \in A$. Question: does the asymptotic density $d(A) = \lim F(N)/N$ exist and satisfy $d(A) > 0$, where $F(N) = |A \cap [1, N]|$? The question is **open**; nothing here settles it. Sections 2-5 establish partial results with proofs, Section 6 reports an exact computational census, and Section 7 derives a heuristic together with its own empirical rejection.

2. Well-foundedness, exact characterization, and generation algorithm

Structural fact [PROVEN]. 1 is not in A (seeds are 2, 3; $ab - 1 \geq 2^2 - 1 = 3$ for $a, b \geq 2$), and whenever $n = a_1 a_2 - 1$ with $a_1, a_2 \in A$ one has $a_1, a_2 \geq 2$ and hence $a_1, a_2 \leq (n+1)/2 < n$. Values strictly decrease along any generation tree toward its leaves.

Theorem 1 (exact characterization) [PROVEN]. n is in A if and only if n is 2 or 3, or $n = ab - 1$ for some a, b in A . *Proof.* Minimality gives one direction directly: if $n > \max(a, b)$ equals $ab - 1$ with $a, b \in A$ then n cannot be excluded by any closed set containing the seeds, induction on $\max(a, b)$ being well-founded by the structural fact. Conversely the set $\{2, 3\} \cup \{ab - 1 : a, b \in A\}$ is contained in A by closure. ■

This makes A the least fixed point of $S \mapsto S \cup \{ab - 1\}$ and justifies the increasing-order dynamic program used by the verifier: membership bits in a bytearray; a min-heap of discovered-but-unexpanded values; a sorted partner list holding $A \cap [1, \sqrt{x}]$. Each popped minimum m is expanded against partners $a \leq \min(m, x/m) \leq \sqrt{x}$.

Theorem 2 (exactness) [PROVEN]. At termination the marked bytes are exactly $A \cap [1, X]$. *Proof.* Soundness is clear: inserted values $ma - 1$ use factors already certified to be in A . For completeness note pushes always exceed the popped value ($ma - 1 \geq 2m - 1 > m$), so pops occur in increasing order; by strong induction on n , before any pop of value $> n$ all elements of $A \cap [1, n]$ are discovered (a non-seed $n' \leq n$ has witness pair with larger factor $M \leq (n'+1)/2 < n$ already discovered and hence expanded while the heap minimum was still below n , inserting n'). Every pair (a, b) with product $\leq X$ is consumed at the expansion of its larger factor $M \leq (X+1)/2$, whose partner $\min(a, b) \leq \min(M, X/M) \leq \sqrt{X}$ is then present in the small list; so expansion against $A \cap [1, \sqrt{X}]$ misses nothing. ■

Cross-check [COMPUTED]. Independent naive fixpoint iteration reproduces the DP census exactly on $[1, 3000]$ ($|A| = 1358$ on both sides). Production run at $X = 10^8$: 6 min 31 s wall time, pure Python standard

library.

3. Lemma A: the modular obstruction and the 2/3 ceiling

Lemma A [PROVEN]. No element of A is congruent to 1 modulo 3. *Proof.* Work in $\mathbb{Z}/3$. The seed residues 2 and 0 both lie in $S = \{0, 2\}$. For r, s in S : if $r = 0$ or $s = 0$ then $rs - 1 = -1 = 2 \pmod{3}$; if $r = s = 2$ then $rs - 1 = 3 = 0 \pmod{3}$. Hence S is closed under the operation and contains the seeds, so by minimality the residues of A lie in S : never 1 modulo 3. ■

Corollary [PROVEN]. Upper density $d^*(A) \leq 2/3 = 0.666667$: each admissible class carries at most $N/3 + O(1)$ elements below N .

Census check [COMPUTED]. Zero of the 55,939,931 elements below 10^8 are 1 mod 3. Note A is a proper subset of the allowed classes: e.g. 6, 11, 62 are not in A although none is 1 mod 3 (their $n+1$ admit no divisor pair from $A \times A$).

4. Residue sets modulo arbitrary moduli

Proposition 3 [PROVEN]. For every modulus m the residue set $\{a \bmod m : a \in A\}$ equals exactly the minimal closed set S_m generated from $\{2 \bmod m, 3 \bmod m\}$. *Proof.* Containment: reduction commutes with the operation, so generation trees reduce into any closed set containing the seed residues. Reverse: every element of S_m arises after finitely many closure steps from the seeds, and each step lifts back through actual elements of A by induction on the step count. ■ Consequently $d^*(A) \leq |S_m|/m$ for every m , with CRT combination across coprime moduli.

modulus m	$ S_m $	$ S_m /m$	remark
3	2	0.666667	$S_3 = \{0, 2\}$: Lemma A
9	6	0.666667	$\{0, 2, 3, 5, 6, 8\}$: repackages mod 3
all other primes $p \leq 500$ (94)	p	1	$S_p = \mathbb{Z}/p$, full set
composite 4..500, m not in $\{9\}$	-	$> 2/3$	none beats 2/3

Table 1. Closed residue sets [COMPUTED]: 95 primes and all composite moduli up to 500 scanned. Only $m = 3$ is proper; the best proven ceiling remains 2/3 and the residue method is saturated at small heights.

5. Lemma B: an explicit logarithmic lower bound

Two families live in A [PROVEN]: $P_k = 2^k + 1$ ($k \geq 1$), the orbit of 3 under $x \mapsto 2x - 1$ (since $t - 1$ doubles along the orbit), and $V_j = (3^{j+1} + 1)/2$ ($j \geq 1$), the orbit of 2 under $x \mapsto 3x - 1$ (fixed point $1/2$: $t_k - 1/2 = 3^k(2 - 1/2)$). Closure produces all pairwise images.

Lemma B [PROVEN]. For $N \geq 8$, $F(N) \geq \text{floor}((\text{floor}(\log_2 N) - 1)^2 / 4) \geq (\log_2 N - 2)^2 / 4 = (\ln N)^2 / (4 \ln^2 2) + O(\ln N)$, with explicit constant $c = 1/(4 \ln^2 2) = 0.5203$.

Proof. Consider $z(i, j) = (2^{i+1} + 1)(2^{j+1} + 1) - 1$ for integers $0 \leq i \leq j$; all lie in A by two applications of closure, and $z = 2^{i+j+2} + 2^{i+1} + 2^{j+1}$. **Distinctness.** If $i < j$ then $z = 2^{i+1}(1 + 2^{j-i} + 2^{j+1})$ with odd bracket, so $v_2(z) = i+1$ and odd part $q = 1 + 2^{j-i} + 2^{j+1}$. If $i = j \geq 1$ then $z = 2^{i+2}(2^{i+1} + 1)$: $v_2(z) = i+2$, odd part $2^i + 1$. The degenerate pair (0,0) gives $z = 8 = 2^3$, the only pure power of two in the family (every off-diagonal entry has odd part $q \geq 11$ and value ≥ 14 ; every diagonal entry with $i \geq 1$ has odd part $2^i + 1 > 1$ and value ≥ 24). Suppose $z(i, j) = z(i', j')$. Off-diagonal vs off-diagonal: equal v_2 forces $i = i'$, and equality of odd parts reads $2^{j-i}(1 + 2^{i+1}) = 2^{j'-i}(1 + 2^{i'+1})$; cancelling the odd integer forces $j = j'$. Diagonal vs diagonal: equal v_2 forces $i = i'$. Off-diagonal vs diagonal would force simultaneously $i+1 = i'+2$ and $1 + 2^{i+1} + 2^{i+2} = 2^{i'+1} + 1$, i.e. $2^{i+1}(1 + 2^{i+1}) = 2^{i'+1}$ with $d \geq 1$: impossible since a pure power of two cannot be divisible by the odd integer $1 + 2^{i+1} \geq 3$. Hence the map is injective. **Counting.** Let $\lambda = \text{floor}(\log_2 N)$; whenever $s := i+j+3 \leq \lambda$, $z \leq 2^{s-1} + 2^{s-2} + 2^{s-2} = 2^s$ with strict inequality (equality would force $s = 2(s-2)+3$), so $z < 2^s \leq N$. The pairs with $i \leq j$, $i+j \leq \lambda-3$ number $\sum_{s=0}^{\lambda-3} \text{floor}(s/2)+1 = \text{floor}((\lambda-1)^2/4)$. ■

Mixed family [COMPUTED; injectivity unproven]. With $U_i = 2^{i+1} + 1$ and $V_j = (3^{j+1} + 1)/2$, the values $w(i, j) = U_i V_j - 1$ all lie in A ; a full injectivity proof would raise the constant to $1/(2 \ln 2 \ln 3) = 0.6566 (\ln N)^2$.

The 2-adic cascade recovers $\min(i+1, v_2(V_j - 1))$ but ties block complete recovery. Computed up to 10^8 : 216 values, zero collisions, all in A; the proven family contributes 168 values against the proven floor 156. Both bounds are exponentially far from the true $F(10^8) = 55,939,931$: their value is unconditional methodology, not approximation.

6. Exact census up to 10^8

k	$F(10^k)$	$F(10^k)/10^k$	local exp	harm incr	delta_hat
1	5	0.500000	-	1.2694	0.5513
2	39	0.390000	0.8921	0.8178	0.3552
3	422	0.422000	1.0342	0.9470	0.4113
4	4,805	0.480500	1.0564	1.1029	0.4790
5	51,508	0.515080	1.0302	1.1838	0.5141
6	535,585	0.535585	1.0170	1.2318	0.5350
7	5,493,428	0.549343	1.0110	1.2637	0.5488
8	55,939,931	0.559399	1.0079	1.2871	0.5590

Table 2. Exact per-decade census [COMPUTED]. local exp = $\log(F(10^k)/F(10^{k-1}))/\log 10$; harm incr = sum of $1/b$ over $A \cap (10^{k-1}, 10^k]$; delta_hat = incr / $\ln 10$ estimates density.

Reading of the data [COMPUTED trend; OPEN conclusion]: raw density rises in every decade; local exponents descend toward 1 from above (top value 1.0079); per-decade harmonic mass approaches delta_hat $\ln 10$ with delta_hat climbing through 0.56. A log-log least-squares fit over decades 4-8 gives $\alpha_{OLS} = 1.01600 \pm 0.00254$ with $R^2 = 0.99998$, consistent with near-linear growth $F(N) \sim c_N N$ and drifting density. All trends are compatible with convergence of $F(N)/N$ to a positive limit $\leq 2/3$; neither existence nor positivity of the limit is provable by these means.

Additional exact facts within the completely stored range $[1, 10^6]$ ($F = 535,585$) [COMPUTED]: first elements 2, 3, 5, 8, 9, 14, 15, 17, 23, 24, 26, ...; smallest non-members 1, 4, 6, 7, 10, 11, 12, 13, 16, 18, ...; maximal gap between consecutive elements is 10 (2159 to 2169); conditional density inside allowed classes $\{n \not\equiv 1 \pmod 3\}$ rises monotonically: 0.591, 0.634, 0.721, 0.773, 0.803 at $N = 10^2, \dots, 10^6$.

7. Mean-field heuristic: derivation and empirical rejection

Membership is an OR over divisor pairs (Theorem 1): n is in A iff $n+1 = ab$ for some a, b in A (or n in $\{2, 3\}$). Assume (H1) equidistribution: for each b , $|A \cap [1, M] \cap (-1 \pmod b)| \sim F(M)/b$ uniformly in b . Counting candidate images $ba - 1 \leq N$ over all b in A gives $F(N) \sim \sum_{b \in A} F(N/b)/b$. Inserting the ansatz $F(N) \sim c_N N^\alpha$ and cancelling yields the self-consistency equation (*): $\sum_{b \in A} b^{-(1+\alpha)} = 1$. Because the truncated harmonic sum diverges like delta_hat $\ln X$ (computed $H_X = 9.1036$ at $X = 10^8$, increments approaching 1.287 per decade), the left side decreases continuously from infinity to zero, so under (H1) a unique $\alpha^* > 0$ exists [HEURISTIC].

Numerics [COMPUTED]. Exact truncated sums below $2 \cdot 10^5$ and 64 logarithmic buckets per decade above give $\alpha^*_{trunc}(10^8) = 0.47344208837849144$ (root residual $+2.5e-11$; stable: 0.472709 already at $X = 10^6$). The integral surrogate $\delta^{2-\alpha} / \alpha = 1$ gives $\alpha^*(\delta = 0.5594) = 0.4185$ and $\alpha^*(\delta = 2/3) = 0.4785$.

Rejection [COMPUTED vs HEURISTIC]. The mean field predicts $F(10^8) \sim (10^8)^{0.4734} = 6,131$ against the exact 55,939,931: off by factor 9,124. Simultaneously the direct fit sits at $\alpha_{OLS} = 1.016 > 1 > \alpha^*$. Two mechanisms identify the failure: (1) *self-affine imaging* - each b in A transports a near-copy $\{ba - 1 : a \in A\}$ into A, so membership correlations inflate density beyond products of marginals (the OR-condition over divisor pairs); (2) *modular rigidity* - equidistribution fails absolutely on class $1 \pmod 3$, and Proposition 3 formalizes such constraints for every modulus. Equation (*) therefore describes a random model of A, not A itself. Its interest is negative but sharp: GREEN-036 is not governed by pseudorandom multiplicative statistics, and any successful approach must exploit the correlated, self-affine structure.

8. Outlook: what is missing

[OPEN] To prove $d(A) > 0$ one wants $F(N) \geq cN$, plausibly via eventual containment of an arithmetic progression of allowed class, or a bootstrap converting lower density at height T into higher density at height $T' > T$. Visible obstructions: 6, 11, 62 are absent despite allowed residues and the gap 2159-2169 shows early holes, so any progression claim needs a threshold phenomenon. To prove $d(A) = 0$ one must reverse a trend rising across eight decades of exact computation; no mechanism suggests itself. Between the two: the only proven ceiling is $2/3$ (Lemma A), the residue method is exhausted for $m \leq 500$, and the proven growth floor is $(\ln N)^2/(4 \ln^2 2)$. Any genuinely new constraint - larger-modulus rigidity, p -adic restrictions, complement stability - would constitute real progress on GREEN-036.

Reproducibility. Everything regenerates by `python3 verify_solution.py [X]` (default $X = 10^7$, about 20 s; production $X = 10^8$, 6.5 min, nine of nine checks passed). Artifacts: `data/census_summary.json`, `data/census.csv.gz` (first 10^6 elements), `data/run_output.txt` reproduced verbatim in the Appendix. Standard library only; deterministic.

The complete dossier is version-controlled at <https://github.com/GamebP/MathProblems/tree/main/GREEN-036>, mirroring this folder file-for-file.

Appendix. Verbatim verifier output (production run, $X = 10^8$)

GREEN-036 verification run, $X = 100000000$

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[1] exact census DP: |A cap [1,100000000]| = 55939931 (351.06s)
[2] naive closure cross-check at X=3000: DP |A|=1358, naive |A|=1358, identical=True (0.57s)
[3] Lemma A: elements congruent to 1 (mod 3) in census: 0 (out of 55939931); predicted 0
    observed raw density  $F(100000000)/100000000 = 0.559399$  (proven upper bound  $2/3 = 0.666667$ )
[4] count table |A cap [1,10^k]| and per-decade increments (25.78s):
    k  F(10^k)  increment  F/N  local exp  harm incr  delta_hat
    1      5        5  0.500000      nan    1.2694  0.5513
    2     39       34  0.390000    0.8921    0.8178  0.3552
    3    422      383  0.422000    1.0342    0.9470  0.4113
    4   4805     4383  0.480500    1.0564    1.1029  0.4790
    5  51508    46703  0.515080    1.0302    1.1838  0.5141
    6 535585   484077  0.535585    1.0170    1.2318  0.5350
    7 5493428  4957843  0.549343    1.0110    1.2637  0.5488
    8 55939931 50446503  0.559399    1.0079    1.2871  0.5590
[5] Lemma B lower-bound families:
    P-orbit  $2^k+1$ ,  $k=1..26$ : 26 values, all in A: True
    Z-products  $(2^{i+1})(2^{j+1})-1 \leq X$  ( $i \leq j$ ): generated 168, in A: 168, collisions: 0
    proven counting formula  $\text{floor}((\text{floor}(\log_2 X)-1)^2/4) = 156$  ( $\leq$  actual 168)
    lower bound achieved:  $168 = 3.00322e-06 \times F(100000000)$ 
    mixed  $(2^{i+1})((3^{j+1})/2)-1 \leq X$ : generated 216, in A: 216, collisions: 0
    asymptotic constants: pure-2 (PROVEN)  $1/(4 \ln^2 2) = 0.52034$ , mixed (if injective)  $1/(2 \ln 2 \ln 3) = 0.65660$ 
    (0.00s)
[6] residue census, 95 primes  $\leq 500$  (6.46s):
    primes with PROPER S_p: [(3, 2)]
    composite moduli 4..500 beating 2/3: none
    S_3 = [0, 2] (Lemma A), |S_p| = p for all other primes  $p \leq 500$ : True
    combined CRT bound over proper prime constraints: 0.666667 (= proven 2/3 here); best single 0.666667
[7] growth-exponent estimates:
    OLS log-log fit over  $k=4..8$ :  $\alpha_{\text{hat}} = 1.01600$  ( $R^2 = 0.999981$ )
    top local exponent  $\ln(F_8/F_7)/\ln 10 = 1.00788$ 
    truncated harmonic sum  $H_X = \sum_{b \in A \text{ cap } [1,100000000]} 1/b = 9.103598$ 
    truncated zeta grid:  $S(0.0)=9.1039$ ,  $S(0.25)=1.8511$ ,  $S(0.5)=0.9483$ ,  $S(0.75)=0.6321$ ,  $S(1.0)=0.4615$ 
    solved  $S_X(\alpha^*) = 1$ :  $\alpha^*_{\text{trunc}} = 0.473442$  (residual  $+2.51e-11$ )
    mean-field prediction  $F(100000000) \sim X^{\alpha^*} = 6131$  vs actual 55939931 (ratio 9124)
[8] artifacts written in 0.9s: census_summary.json, census.csv.gz (first 1000000 elements) sizes: 3618B, 2285592B
=====
[PASS] EXACTNESS      DP set == naive fixpoint closure on [1,3000] (|A| = 1358 both sides: True)
[PASS] LEMMA-A        0 violations of  $n \equiv 1 \pmod 3$  among 55939931 census elements
[PASS] PORBIT         all 26 values  $2^k+1 \leq X$  lie in A
[PASS] LEMMA-B-Z      168/168 Z-products in A, 0 collisions, proven formula  $156 \leq$  actual
[PASS] LEMMA-B-MIXED  216/216 mixed products in A, 0 collisions (injectivity COMPUTED, not proven)
[PASS] RESIDUES       S_3 = {0,2}; proper-prime list [(3, 2)]; combined bound 0.666667  $\geq$  observed density 0.559399
[PASS] MONOTONE       decade counts strictly increasing, e.g. 5 -> 55939931
[PASS] ZETA-ROOT      S_X( $\alpha^*$ ) = 1 at  $\alpha^* = 0.473442$ , residual  $+2.51e-11$ 
[PASS] DENSITY-TREND  F/N still rising: 0.5493 -> 0.5594 (question OPEN)
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```

9/9 checks passed. Problem GREEN-036 remains OPEN; this run establishes computational evidence only.